

Passive Microwave Sea Ice Concentration

Introduction

Brightness temperature (TB) data from passive microwave radiometers are widely used to estimate sea ice concentration. The objective of the present assignment is to investigate brightness temperature data, and estimate sea ice concentration from these data.

Background Reading material

Ivanova, N. et al. Inter-comparison and evaluation of sea ice algorithms: towards further identification of challenges and optimal approach using passive microwave observations, The Cryosphere, 9, 1797-1817, 2015 (available online)

You will be provided with 'swath' data of brightness temperatures, representing satellite observational coverage for January, 24, 2014. The brightness temperature file is:

[GW1AM2_201401240631_012D_L1SGRTBR_2210210.nc](#)

From this file you will extract data for the in the Gulf of St. Lawrence (45°N - 52°N, 70°W - 58°W), a region along the Canadian East coast with seasonal sea ice cover . In this region, the ice melts completely each year, hence there is no multi-year ice. The winter of 2014 was very cold, and these data represent the period during which the ice was forming and growing thicker.

Study domain, Gulf of Saint Lawrence, shown as boxed region.



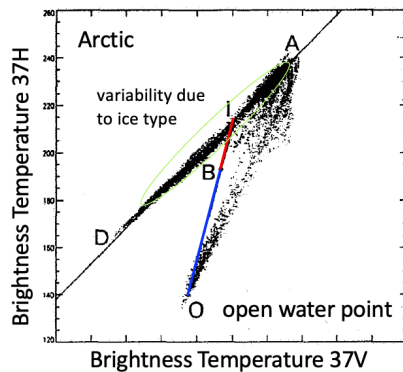
The assignment is comprised of the following steps. For all steps use only data extracted over the Gulf of Saint Lawrence (GSL)

1. Write a program to read the data (in netCDF format) and extract points for the Gulf of Saint Lawrence (GSL). We will use the following variables from the netcdf file:

Number	Variable Name	Description
58	lat	latitude
59	lon	longitude

14	Brightness_Temperature__res06_6_9GHz_H_	6.9 GHz brightness temperature, horizontal polarization
15	Brightness_Temperature__res06_6_9GHz_V_	6.9 GHz brightness temperature, vertical polarization
38	Brightness_Temperature__res36_36_5GHz_H_	36.5 GHz brightness temperature, horizontal polarization
39	Brightness_Temperature__res36_36_5GHz_V_	36.5 GHz brightness temperature, vertical polarization
44	Land_Ocean_Flag_6_to_36	Land ocean flag

2. Use the 'Land_Ocean_Flag_6_to_36' to remove land points for 6.9 GHz and 36.5 GHz. The first matrix in 'Land_Ocean_Flag6' is for 6.9GHz, the second one is for 10.7 GHz, the third one is for 18.7 GHz (sometimes called 19 GHz) whereas the fourth one is for 36.5 GHz. To remove the land points set the brightness temperatures (BT) to NaN for points where the Land_Ocean_Flag value is greater than 50%. Do this for 6.9GHz and 36.5 GHz separately.
3. Produce plots of brightness temperature (BT). Show spatial maps of BT values for 6.9 GHz (horizontal polarization) and 36.5 GHz horizontal polarization) with land points removed (see step 2) for the GSL. You will need to convert the datatype for brightness temperature so it is a real number, and multiply by a scaling factor (which is 0.01) that brings it to units of Kelvin.
4. Create an x-y scatter plot of the two different polarizations at 36.5 GHz (these are shown as 37 GHz in the Comiso paper because at that time the channel was centered at 37 GHz) . You may wish to flatten the data to be a vector before doing this. Does the distribution of your points look similar to those in Comiso (Fig 3) shown below? Write a sentence or two describing why or why not.



5. We will now do a single channel retrieval using the 36H channel. To do this start by plotting a histogram of the BT data for BT_36H, be sure to include only non-land points. You should see something like the figure below:

The points corresponding to low values of brightness temperature are open water, while those corresponding to high values are high concentration sea ice (recall the ice is radiometrically warm relative to water, so it has a higher brightness temperature). To obtain a water tie point, use the value corresponding to the 5th percentile of BT_36H, in Matlab this is done using the `prctile` function, for example denoting the water tie point by T_W , $T_W = \text{prctile}(\text{BT_36H}, 5)$. To obtain an ice tie point follow a similar process, using instead the 95th percentile of BT_36H. For example denoting the ice tie point by T_I , $T_I = \text{prctile}(\text{BT_36H}, 95)$

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6. With these tie points, estimate sea ice concentration using a linear mixture model for each of the given days and each of your chosen frequency and/or polarization combinations. Recall in the linear mixture model the sea ice concentration (C) is calculated from,

$$C = \frac{BT - T_W}{T_I - T_W}$$

where BT is the measured brightness temperature, and T_W , T_I are the water and ice tie points. Apply this equation to each non-land measurement location in the GSL, where T_B the 36.5 H brightness temperature at the non-land data point.

Plot your sea ice concentration estimates for each day and comment on what you see. Note that you may have sea ice concentration estimates that are greater than one or negative.

7. Now we will compare your estimated sea ice concentrations with data from a data provider. A data file for sea ice concentration is provided, with a script to plot the sea ice concentration. The sea ice concentration filename is:

ESACCI-SEAICE-L4-SICONC-AMSR_25.0kmEASE2-NH-20140131-fv2.0

For reference, the site the data were downloaded from is:

https://data.ceda.ac.uk/neodc/esacci/sea_ice/data/sea_ice_concentration/L4/amsr/25km/v2.0/NH/2014/01

Documentation about the dataset is here:

https://dap.ceda.ac.uk/neodc/esacci/sea_ice/docs/SICCI_P2_SIC_PUG_D3.3_Issue_1.0.pdf

Using January 24th as an example, plot the sea ice concentration from the data provider. Compare this sea ice concentration to one of the ones you've estimated, commenting on any differences.